Electrostatic Shielding and Hydrogen Evolution Suppression via Tetraethylammonium Bromide in Rechargeable Zinc-Ion Batteries

Authors: Alex Mercer, Bowen Liu, Elena Rostova
Affiliation: National Research Council Canada & University of Alberta
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Abstract: Aqueous zinc-ion batteries (ZIBs) are safe, eco-friendly energy storage devices, but their practical implementation is impeded by zinc dendrite growth and parasitic hydrogen evolution reaction (HER). Herein, we report that tetraethylammonium bromide (TEABr) acts as a multifunctional electrolyte additive in a 2 M ZnSO4 baseline aqueous electrolyte. Through systematic optimization (10 mM, 25 mM, 50 mM, 100 mM), an optimal concentration of 50 mM TEABr was identified. The bulky quaternary ammonium TEA+ cations self-assemble at localized high-electric-field protuberances, establishing a temporary electrostatic shield that prevents localized Zn²■ accumulation. Simultaneously, bromide anions modify the interfacial adsorption layer, yielding an exceptionally smooth zinc plating morphology with preferred (002) texture. Consequently, Zn//Zn symmetric cells with 50 mM TEABr deliver an outstanding lifespan of 1650 hours at 2.0 mA cm-2 and 1.0 mAh cm-2 with an overpotential of 32 mV. Zn//Cu asymmetric cells exhibit a high average Coulombic efficiency of 99.5% over 800 cycles.

Electrochemical Evaluation: All electrochemical cycling experiments were performed in 2032 coin-type cells using polished zinc foil (100 µm thickness) as both working and counter electrodes. The baseline electrolyte was 2 M ZnSO4 dissolved in deionized water (H2O). The additive tetraethylammonium bromide (TEABr, chemical formula C8H20BrN, SMILES: CC[N+](CC)(CC)CC.[Br-]) was dissolved directly into the baseline electrolyte at 50 mM without any co-solvent. Linear sweep voltammetry (LSV) confirmed that 50 mM TEABr increased the hydrogen evolution overpotential by 115 mV compared to the bare 2 M ZnSO4 electrolyte.